



Time is in the head

The story of how spending time in orbit caused astronomer Scott Kelly to age slower than his twin brother, Mark. <https://digit.in/apr20-65>



The Nature of the Physical World

Read the entire book by Sir Arthur Stanley Eddington, that proposed the concept of the arrow of time. <https://digit.in/apr20-66>

again, it is intuitive to think that the same amount of time has passed for both of them. However, according to the theory of relativity, time actually passes slower for the astronaut than his grandfather, simply because the grandfather was resting while the astronaut was in motion. In fact, we have actually seen this happen, the astronaut Scott Kelly aged slightly slower than his twin brother Mark Kelly. Essentially, the faster you travel, the slower time passes for you. Extreme examples of this kind of time warping, known as time dilation, was seen in the movie *Interstellar*. We all experience time slightly differently depending on how fast we are moving, so if you travel a lot, you actually live longer.

ENTROPY

For the most part, the question of the exact reason why time has a direction, boils down to entropy, which related to the second law of thermodynamics. There are many complicated statements of the law, but essentially it means that the universe goes from order to disorder, which is true for small isolated systems as well. For example, a hot pizza cools down when exposed in a room, a process that can be slowed down by keeping it in a box. On a universal scale, this can be thought of as the tendency for everything, that is both matter and energy to disperse. Although there are some local exceptions, such as stars forming from diffuse clouds of matter, by and large the universe is getting bigger, and colder. The energy concentrated in the stars and galaxies are slowly dissipating.

Ludwig Boltzmann explained thermal conductivity by using statistics, and suggested that the second law of thermodynamics was just a statistical function, in the sense that the disorder was simply more probable than order. For example, while it is possible to shuffle a deck of cards in such a way that it results in the order that they are packed in a new box, it is just extremely improbable. However, Sir Alfred Brian Pippard, a British physicist argued against the notion by stating

that “there is thus no justification for the view, often glibly repeated, that the Second Law of Thermodynamics is only statistically true, in the sense that microscopic violations repeatedly occur, but never violations of any serious magnitude. On the contrary, no evidence has ever been presented that the Second Law breaks down under any circumstances.” However, this is not entirely true, as there are some theoretical situations where the second law of thermodynamics does not hold true, and this is as yet an unresolved issue.



Mark Kelly (left) aged five milliseconds older than his twin brother Scott Kelly (right) while the latter spent a year in space

QUANTUM MECHANICS

At very small scales, things start becoming a little weird. Scientists have not been able to establish a link between the thermodynamic arrow of time and the quantum arrow of time, which is a description of how very small objects change over time. One of the problematic aspects is at the heart of what quantum mechanics is. There is a point of view, which is very common and taught in schools, called the Copenhagen Interpretation. The pertinent aspect about it is that quantum mechanics only describe the physical systems in terms of probabilities. For example, it is possible to say that an electron exists in such energy levels around the nucleus of an atom. According to the Copenhagen Interpretation, there is no real position and velocity of

the electron around the atom, it exists in all the possible positions at once, until it is observed or measured. When the electron is observed, what follows is called a wave function collapse, after which certain aspects of the electron can be measured, such as its position or its velocity. Now an observer, as in a human is subjective, something that should not be the case with physics, and it is difficult to define a measuring device, so the very idea of a wave function collapse is a little obtuse. Over and above that, the exact mechanism

through which such a wave function collapse occurs is little understood.

There are also multiple exciting new theories, including one by researcher, Robert Lanza, which is a biocentric view of the cosmos. These theories are controversial, and go a step further than Einstein's theory of relativity. Instead of time being dependent on the observer, the observer actually creates time. According to Lanza, space and time are not physical things, but “forms of animal intuition”, or as Kant explained it, modes of understanding. Another scientist, Carlo Rovelli also suggests that the physicality of time is intrinsically linked to our perception of it. According to Rovelli, the direction of time is an illusion, one that can be compared to the apparent rotation of the sky at night. 